

MATTHEW YU

Software Engineer · AI systems, cloud infrastructure and full-stack development
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PROFESSIONAL SUMMARY

Software developer with a Master of Information Technology from QUT and 2 years of professional engineering experience at Delta Electronics. Skilled in Python, JavaScript, TypeScript, and React/React Native, with hands-on experience across AI/LLM integration, full-stack web and mobile development, and cyber security. 1st place winner, 2026 QUT Sport Innovation Hackathon. Currently building an AI-powered vulnerability prioritisation platform as an industry capstone project; shipped AnyWear, a peer-to-peer travel wardrobe app with 20K+ users on the App Store.

TECHNICAL SKILLS

Languages: Python, C#, JavaScript, TypeScript, Go, SQL, HTML/CSS
Frameworks & Libraries: React, React Native, Expo, Next.js, Node.js, Flask, Bootstrap, Tailwind CSS, scikit-learn, Pandas, NumPy
Databases: PostgreSQL, Supabase, MongoDB, SQLite, MySQL
AI / ML: Claude API, Ollama (local LLM), RAG pipelines, agent design, prompt engineering, scikit-learn, supervised/unsupervised learning
Cloud & Tools: AWS (EC2, S3, RDS, VPC, IAM), Terraform, Docker, Git, GitHub Actions (CI/CD), Vercel, Expo EAS, Jupyter Notebook
Concepts: Machine Learning, Agile/Scrum, SDLC, Cyber Security, Network Security, REST APIs, OOP, Human-Centred Design, IoT

PROJECTS

-  **QUT Sport Innovation Hackathon — 1st Place**

Apr 2026

Python, Flask, Claude API | Team of 5 | Team SightLine

 - Brisbane's local sport engagement was projected to spike during the 2032 Olympics and drop sharply afterward; the team was tasked to design for lasting fan engagement.
 - Built "Educated Spectating", an AI platform using Python and Flask integrated with the Claude API to generate real-time, beginner-friendly explanations of rules, tactics and key moments during live sports broadcasts.
 - Won 1st place among all entrants and secured selection for an 8-week MVP development program with Innovation Central Brisbane, backed by QUT Sport as part of the Brisbane 2032 legacy initiative.
- AI-Powered Vulnerability Management Platform (Vertex) — QUT Industry Project**

2026

Vite, React, TypeScript, Go, PostgreSQL, Ollama, RAG, Terraform, AWS | Client: QUT Information Security

 - Existing vulnerability tools rank findings without organisational context, forcing security teams to manually triage which incidents matter most.
 - Co-developing a prioritisation platform with a Go backend, React/TypeScript frontend and PostgreSQL data layer, using a RAG pipeline and a self-hosted Ollama model to layer client-specific context onto a deterministic scoring engine while keeping sensitive data off third-party infrastructure.
 - Independently own cloud deployment via Terraform on AWS (EC2, S3, RDS, VPC, IAM); delivering a working prototype for a real client despite limited access to sensitive validation data.
- AnyWear — Peer-to-Peer Travel Wardrobe App (iOS/Android)**

2026

React Native, Expo, Supabase (PostgreSQL, RLS), Stripe Connect, Google Maps API

 - Built a full-stack cross-platform mobile app single-handedly — listings with photo upload, bookings, Stripe Connect payments, real-time messaging, reviews and ID verification — while iterating features alongside an evolving business plan.

- Shipped to 20K+ users on the App Store; building through business-model changes reinforced the value of locking down system design and business logic before implementation, an approach now applied on Vertex.

VIREX — E-Commerce AI Automation Agent

2026

Node.js, REST API, Claude Sonnet

- Designed and built an AI agent with a Node.js/REST API backend, integrating Claude to generate and publish product listing content end to end.
- Delivered a working automation pipeline; identified that loosely specified prompts produced unstable output, sharpening a practical skill in precisely scoping LLM output requirements now applied across every AI-integration project.

Coursework projects: Online Recruitment & Management Application (IFN636 — React, Tailwind CSS, Node.js, MongoDB, GitHub Actions CI/CD) · Online Food Delivery Website (IFN582 — Bootstrap, Python/Flask, MySQL) · Cloud Computing (CAB432 — Terraform, AWS EC2/S3/RDS/VPC/IAM).

PROFESSIONAL EXPERIENCE

Mechanical Engineer | Delta Electronics

Apr 2023 – Feb 2025

Taoyuan, Taiwan

- Collaborated with Microsoft engineering teams on cooling system design for Azure data centres, contributing to thermal performance and energy efficiency improvements through iterative design and field-test validation.
- Led problem analysis and root-cause investigation for engineering changes and sample production issues, supporting design decisions across multiple product lines.
- Managed Product Bill of Materials (BOM) drafting and verification, coordinating part material selection and supplier qualification across cross-functional design, manufacturing and QA teams.

EDUCATION

Master of Information Technology

Feb 2025 – Dec 2026

Queensland University of Technology — Brisbane, QLD
Machine Learning, Cloud Computing, Advanced Algorithms and Complexity, Rapid Web Development with Databases, Software Life Cycle Management, Cyber Security and Governance, Networks and Security, Human-Centred Design of IT Systems, Industry Project.

B.Sc. Mechanical Engineering

Jun 2022

Yuan Ze University — Taoyuan, Taiwan

AWARDS

- 1st Place — 2026 QUT Sport Innovation Hackathon (Team SightLine)
- Undergraduate Thesis Excellence Award — Yuan Ze University
- Cross-Culture Communication Award — Yuan Ze University

LANGUAGES

- Mandarin Chinese (native)
- English (professional working proficiency)